

WEEK 6



Building a Well

Build up, then dig down

벽을 쌓고 아래로 파기

BIG PICTURE

Two skills in one build



A square wall on top + a hole dug down below.

사각 벽 + 깊은 구멍

HOW TO COUNT

Count the axis — one block, one number



- 1 Stand on the start block — that spot is $(0, 0, 0)$.
- 2 $x \rightarrow$ walk **right**. Add **+1** for every block.
- 3 $y \rightarrow$ jump **up**. Add **+1** for every block.
- 4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

A square = 4 sides + 4 turns

```
repeat 4 times:  
  repeat 3 times:  
    place block down  
    move forward  
  turn right
```

Build a side, turn right, repeat **4 times** → a closed square.

정사각형 = 4면 + 4번 회전

CONCEPT

Digging = destroy + go down

```
repeat 3 times:  
  destroy block down  
  move down by 1
```

`destroy block down` breaks the block, then the agent drops into the gap.

파기 = 부수고 + 내려가기

CONCEPT

place vs destroy



place block

adds a block



destroy block

removes a block

놓기와 부수기는 반대예요

CODE

Blocks: wall first, then dig

```
# 1) the square wall
repeat 4 times:
    repeat 3 times:
        place block down
        move forward
    turn right
# 2) dig the hole
repeat 3 times:
    destroy block down
    move down by 1
```


CODE

How the teacher world builds it

```
# well - week 6 result  
blocks.fill(COBBLESTONE, pos(20,0,0), pos(24,1,4))  
blocks.fill(AIR, pos(21,0,1), pos(23,1,3))  
blocks.fill(WATER, pos(21,0,1), pos(23,0,3))
```

Fill cobblestone, carve out **AIR** for the hole, then add **WATER**.

AIR 로 비우고 WATER 로 채워요

PREDICT 🤖

How deep is the hole?

Predict first — then click reveal

```
repeat 3 times:  
  destroy block down  
  move down by 1
```

Reveal output 🔍 reveal

VOCAB

New words today

destroy break and remove a block

dig remove blocks going downward

deep how far down something goes

square a shape with 4 equal sides

부수다

파다

깊은

정사각형



What's wrong with this code?

Goal: **dig 3 blocks straight down.**

buggy

```
repeat 3 times:  
  destroy block down
```

Think first — then click next.



DEBUG

The bug

Bug: the agent breaks the same block 3 times without moving — it never goes deeper. Add `move down by 1` so it drops into the hole each loop.

fixed

```
repeat 3 times:  
  destroy block down  
  move down by 1
```

YOUR TURN!

Build a deeper well



Open this week's world



Build a **3×3 square wall** with a loop



Dig the hole **at least 5 blocks deep**



Build the wall *first*, then dig



Send a photo looking down into your well



QUIZ

QUESTION 1 OF 3

How many times must you repeat "build a side, turn right" for a square?

10 times.

4 times.

2 times.

1 time.



QUIZ

QUESTION 2 OF 3

Why must digging include `move down by 1`?

To place water.

So the agent drops deeper each loop.

It isn't needed.

To turn the agent around.



QUIZ

QUESTION 3 OF 3

destroy block down does the opposite of:

move forward

place block down

turn right

repeat

QUIZ

Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



Well dug!

Next: a farm house with variables.

다음 주: 변수로 농장 집 짓기